

01-July-2026

Agenda:

ML-13

→ support Vector Machine (svm)

→ primal form (linear svm) → completed

$$J(w, b) = \frac{1}{2} \|w\|^2 + c \sum_{i=1}^n \max(0, 1 - y_i (w \cdot x_i + b))$$

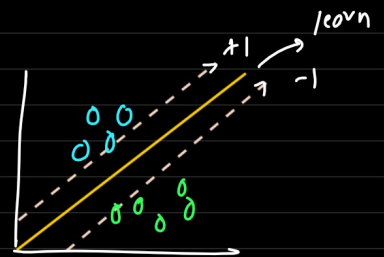
path A: $(z \geq 1)$, loss $\rightarrow 0$

$$\frac{\partial J}{\partial w} = w \quad \frac{\partial J}{\partial b} = 0$$

path B: $(z < 1)$, loss is active

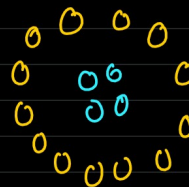
$$\frac{\partial J}{\partial w} = w - (c \cdot y_i \cdot x_i)$$

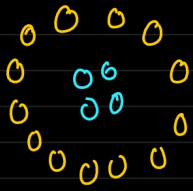
$$\frac{\partial J}{\partial b} = -(c \cdot y_i)$$



The problem: when straight line fails

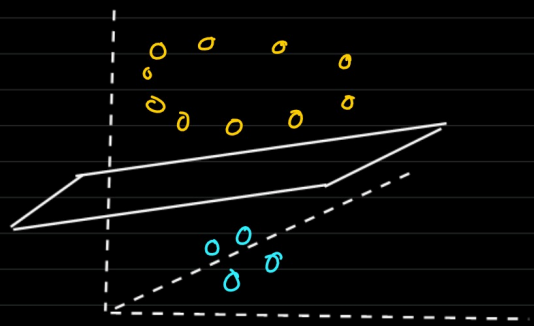
linear regression $\rightarrow$ polynomial regression





2D

transform the data
into higher dimension



(Dual form - kernelized SVM)

$$A: [2, 3] \rightarrow$$

2D

(polynomial function)

$$\phi(x) =$$

$$d=2$$

$$[x_1^2, \sqrt{2}x_1x_2, x_2^2]$$

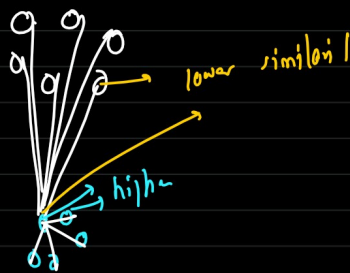
$$A: [2^2, \sqrt{2(2)(3)}, 3^2]$$

$$[4, 6\sqrt{2}, 9]$$

3D

$$B: [4, 1]$$

$$[16, 4\sqrt{2}, 1]$$



Requirement:

1) mathematical function $\rightarrow$ high dimensional data

2) close or similar to their class $\rightarrow$ dot product

$$A: [4, 6\sqrt{2}, 9] \quad B: [16, 4\sqrt{2}, 1]$$

→ Take the 3D dot product:

$$(4 \times 16) + (6\sqrt{2} \times 4\sqrt{2}) + (9 \times 1) = 64 + (24 \times 2) + 9 \\ = 121$$

2D → 3D → dot product → single value ← similarity score

A-B

closer → higher score

far → lower score

2D → 3D

100,000

3D

□ □ □

→ memory
→ computation

100,000 × 3

computation

dot product score

$$\rightarrow (A \cdot B)^2$$

step-1 → A → 3D

step-2 → B → 3D

step-3 → A dot B

$$\left((2,3) \cdot (4,1) \right)^2 \rightarrow \left((2 \times 4) + (3 \times 1) \right)^2 \rightarrow \left((8) + (3) \right)^2 \\ \rightarrow (11)^2 \rightarrow 121$$

kernel
function

or

kernel trick

$$A - B \rightarrow \square$$

$$A - C \rightarrow \square$$

computer

3D

$$\text{car} \rightarrow (\text{red}, 240 \text{ km/hr}, 40) \rightarrow A$$

$$\text{car} \rightarrow (\text{yellow}, 120 \text{ km/hr}, 20) \rightarrow B$$

$$\text{car} \rightarrow (\text{yellow}, 230 \text{ km/hr}, 35) \rightarrow C$$

$$\text{car} \rightarrow (\text{blue}, 110 \text{ km/hr}, 5) \rightarrow D$$

$$\text{car} \rightarrow (\text{green}, 130 \text{ km/hr}, 10) \rightarrow \square$$

3D

B, C

2D $\rightarrow$ ND

linear kernel:

SVM $\rightarrow$ kernel $\rightarrow$ linear

$$\phi(x) = x$$

$$\phi(2,3) = [2, 3]$$

$$\phi(2,3)$$

$$\phi(4,1)$$

A · B

polynomial kernel:

$$\phi(x) : [x_1^2, \sqrt{2}x_1x_2, x_2^2]$$

$d=2$

$$\phi(2,3) : [4, 6\sqrt{2}, 9] \rightarrow (A \cdot B)^2$$

$$\phi(4,1) : [16, 4\sqrt{2}, 1]$$

RBF kernel (Radial Basis function / Gaussian)

maps data to infinite dimension

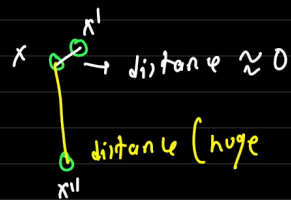
$\rightarrow$ Taylor series $\rightarrow e^x \rightarrow$ polynomial goes forever

$$k(x, x') = \exp(-\gamma \|x - x'\|^2)$$

$\rightarrow$ square euclidean distance between two point

data $\rightarrow$ infinite $\rightarrow$ kernel $\rightarrow$

instead of thinking about infinite dimension, think about close & far points are



point right on top of each other
→ distance 0

→ distance 0

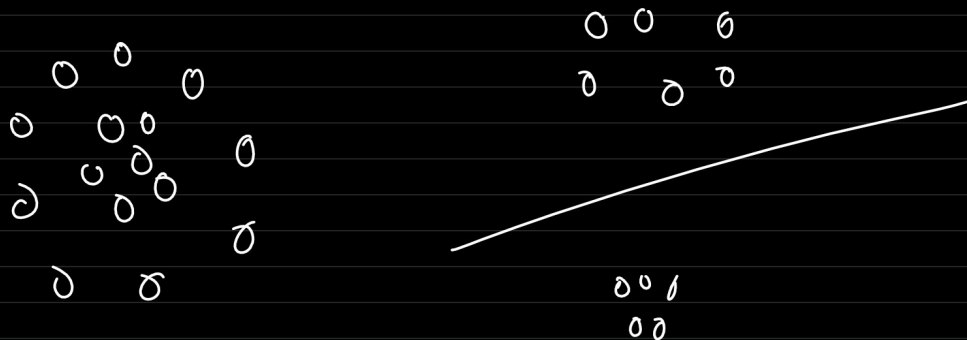
$$\rightarrow \exp(0) \rightarrow 1$$

point that are miles apart

→ distance huge

$$\rightarrow \exp(-h\nu_e) \approx 0$$

RBF $\rightarrow$ similarity score ranging from 1 (identical) to 0 (unrelated)



→ 10 → 20 → 5 → 9
Hello, How are you → ① [10, 20, 5, 9]

Hallo, are you How $\rightarrow$ (2) $[10, 5, 4, 20]$

Hello $\rightarrow$ (0 0 0 0 0 0 0) $\xrightarrow{1 \times 512}$
 Hello. $\rightarrow$ (0 0 0 0 0 0 0) $\xrightarrow{1 \times 512}$